



SECJ3623

MOBILE APPLICATION PROJECT

FINAL PROJECT REPORT

INVENTORY MANAGEMENT SYSTEM

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Introduction

Overview

The Inventory Management System (IMS) is a mobile-based application developed to support and enhance the management of spare parts and inventory operations within the engineering store at FNBN. The system is designed as a digital solution to overcome inefficiencies caused by manual and semi-digital processes, ensuring better accuracy, traceability, and operational efficiency.

This report documents the complete analysis, design, and prototype development of the IMS, focusing on system requirements, user needs, and logical workflows.

Project Background

Engineering store operations at FNBN currently depend on manual stock cards, Excel spreadsheets, and paper-based documentation. While these methods are widely used, they introduce several operational challenges such as delayed stock updates, inconsistent records, difficulty in tracing spare part usage, and an increased risk of human error.

In maintenance-driven environments where machine uptime is critical, such limitations can directly impact productivity. The IMS project is proposed to digitalize inventory operations and provide a centralized platform that supports real-time tracking and informed decision-making.

Project Objectives

The primary objectives of the IMS project are as follows:

- To eliminate manual and paper-based inventory management processes.
- To provide real-time visibility of inventory levels and storage locations.
- To ensure complete traceability of spare parts linked to work orders and assets.
- To reduce machine downtime through faster and more accurate spare part issuance.
- To support proactive procurement using alerts and historical usage data.

Stakeholders and Users

Identified Stakeholders

The IMS project involves multiple stakeholders who influence or benefit from the system:

- Engineering Department
- Maintenance Team
- Store Management
- FNBN Operations Management

System Users

The system is designed for two primary user roles:

Admin / Store Supervisor

Responsible for system configuration, inventory control, purchase order management, and report generation.

Storekeeper / Technician

Responsible for issuing, receiving, and consuming spare parts during maintenance and operational activities.

Requirement Analysis

Overview

The Requirement Analysis phase focuses on identifying, analyzing, and documenting the functional and non-functional requirements of the IMS. These requirements are derived from existing operational challenges, stakeholder discussions, and the approved project proposal.

The objective of this phase is to ensure that the system effectively replaces manual processes while supporting accuracy, traceability, and reporting needs.

Existing System Analysis

The current inventory management approach at FNBN relies heavily on manual stock cards, Excel files, and paper-based documentation. These methods result in several operational issues, including delayed and inconsistent stock updates, difficulty in tracking spare part history, and limited real-time visibility of inventory levels.

Additionally, manual processes increase the risk of human error, data duplication, and inefficiencies during reporting and auditing. These limitations highlight the need for a centralized and automated inventory management system.

Proposed System Scope

The Inventory Management System (IMS) will be developed as a mobile application to centralize and automate inventory-related operations. The scope of the proposed system includes:

- User authentication with role-based access control
- Inventory master data management
- Inventory consumption and issuance tracking
- Purchase order management
- Alerts and reporting functionalities

Functional Requirements

User Management

The system shall allow users to log in using authenticated credentials. Role-based access control shall be implemented to restrict system features based on user responsibilities.

Inventory Management

Authorized users shall be able to create, update, view, and delete item records. The system shall display real-time stock levels and storage locations while maintaining a centralized master list of spare parts.

Inventory Consumption and Issuance

The system shall support barcode scanning for issuing and receiving spare parts. Inventory quantities shall be updated automatically after each transaction, and issued parts shall be linked to specific work orders or assets. A complete transaction history shall be maintained for traceability.

Purchase Order Management

The system shall allow users to create and track purchase orders. Stock levels shall be updated automatically upon successful receipt of ordered items.

Alerts and Reporting

The system shall generate low-stock and critical spare alerts. Administrators shall be able to generate inventory and usage reports and review historical data for audit purposes.

Non-Functional Requirements

Performance

The system shall support real-time inventory updates and concurrent user access without data inconsistency.

Usability

The system shall provide a user-friendly mobile interface and minimize manual input through barcode scanning and predefined selections.

Security

Secure authentication and authorization mechanisms shall be enforced to prevent unauthorized access to sensitive data.

Reliability

The system shall ensure data integrity during transactions and maintain accurate system logs for all actions performed.

System Design Overview

The IMS follows a mobile-client and cloud-backend architecture. The mobile application handles user interaction and barcode scanning, while the backend manages authentication, data storage, and synchronization.

Below are the complete UML diagrams for this system to represent system structure and user interactions. Additionally added a system architecture design. To view the entire UML diagram outlined below, attached the link below:

[Inventory Management System UML Diagram](#)

User Register/Login & Main Page

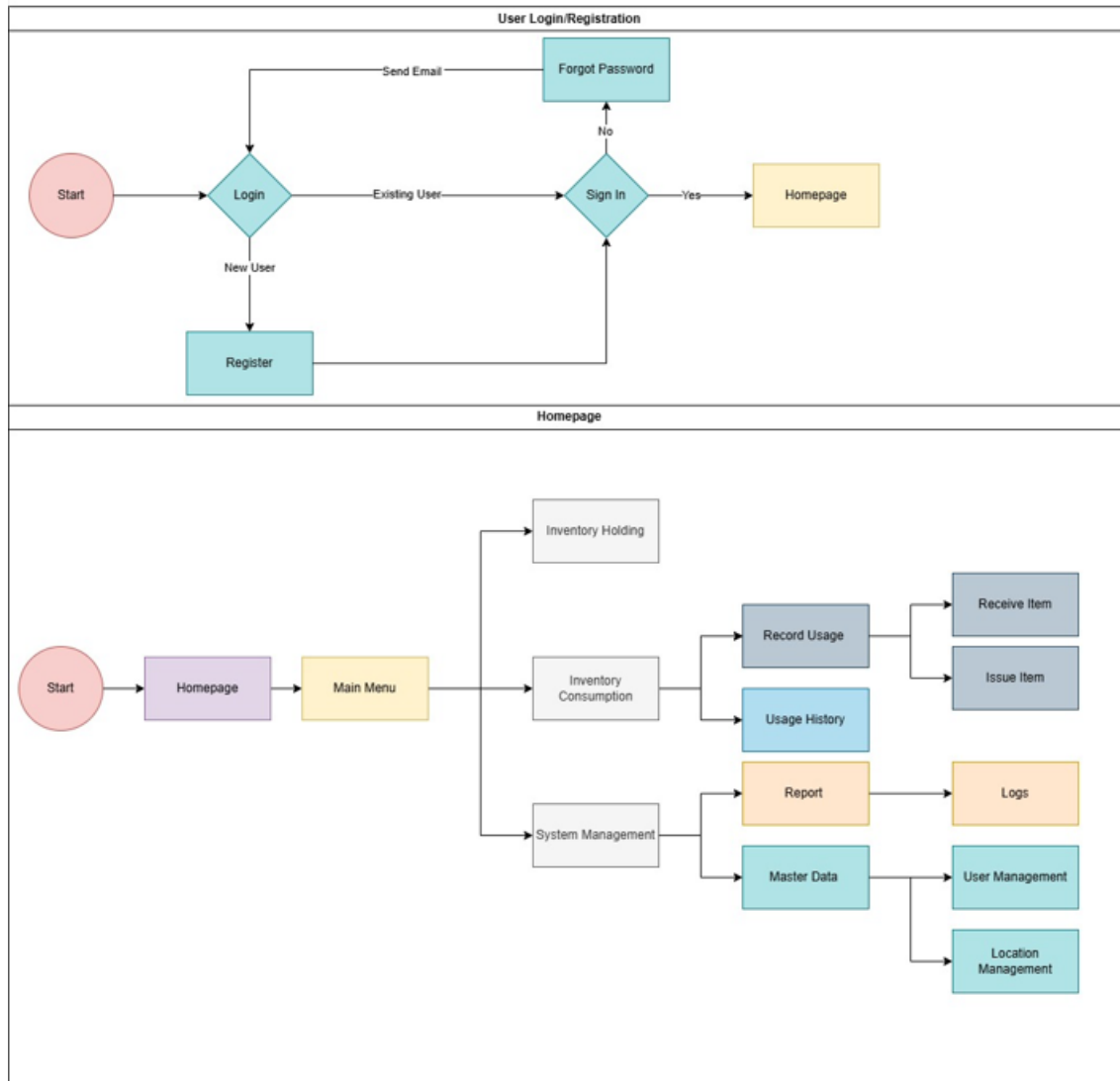


Image 1: UML diagram for user register/login and main page

Inventory Management Modules

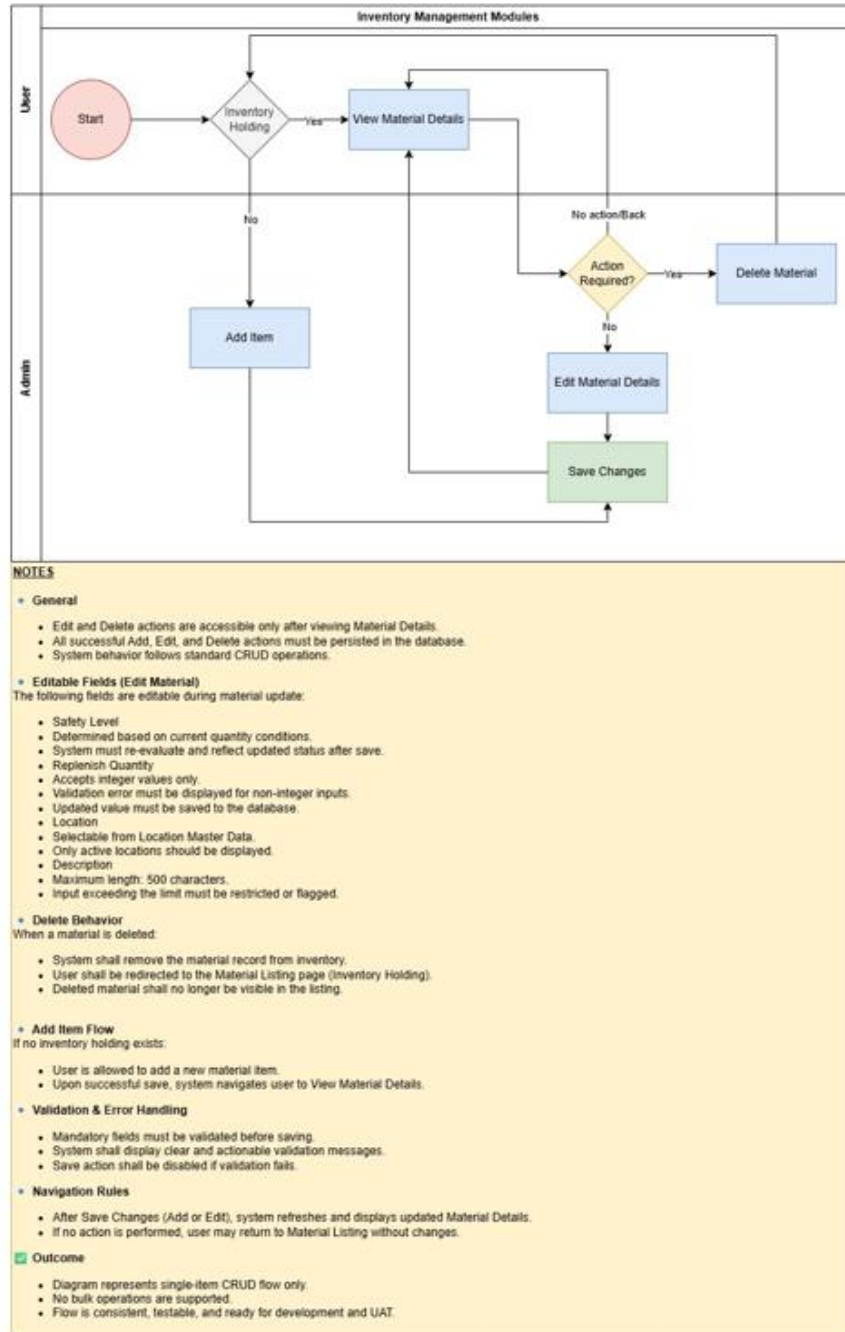
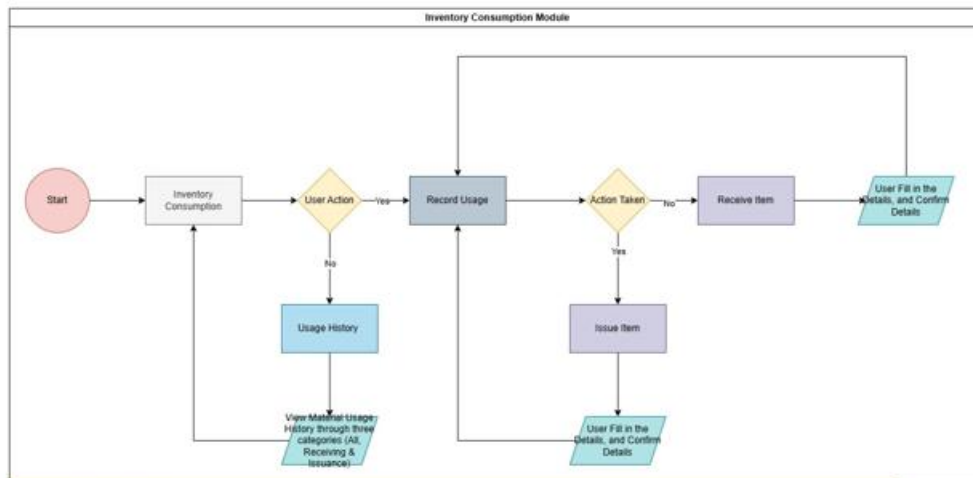


Image 2: UML diagram for inventory managements modules

Inventory Consumption Module



Flow Explanation (Step by Step)

1. Start

- The process begins when a user accesses the Inventory Consumption Module

2. Inventory Consumption

- The system opens the inventory consumption screen where users can:
 - Record usage of items
 - Issue item
 - Receive items
 - View usage history

3. User Action Decision

- The system checks what action the user wants to perform.
- If NO action is selected:
 - The system directs the user to Usage History
- If YES (an action is selected):
 - The system proceeds to Record Usage

4. Usage History

- Users can view past inventory records.
- Displays material usage history
- Categorized into:
 - All records
 - Receiving
 - Issuance
- This allows users to track and review inventory movement.

5. Record Usage

- The system prepares to record a new inventory transaction.
- All inventory changes must pass through this step
- Ensures data consistency and traceability

6. Action Taken Decision

- The system checks what type of action is being recorded.
- If YES (item is being issued):
 - Proceed to Issue Item
- If NO (item is being received):
 - Proceed to Receive Item

7. Issue Item

- Used when inventory items are given out or consumed.
- User enters required details
- Confirms the transaction
- Quantity is deducted from inventory

8. Receive Item

- Used when inventory items are received or returned.
- User enters required details
- Confirms the transaction
- Quantity is added to inventory

9. User Fills Details & Confirms

- For both Issue and Receive:
 - User fills in item details (Quantity, Usage Location & Technician Name.)
 - User confirms the action
 - System saves the record into usage history
 - End Result
 - Inventory levels are updated correctly
 - All actions are recorded in usage history
 - Users can track all inventory movements clearly
 - Reduces errors and improves accountability

Image 3: UML diagram for Inventory consumption modules

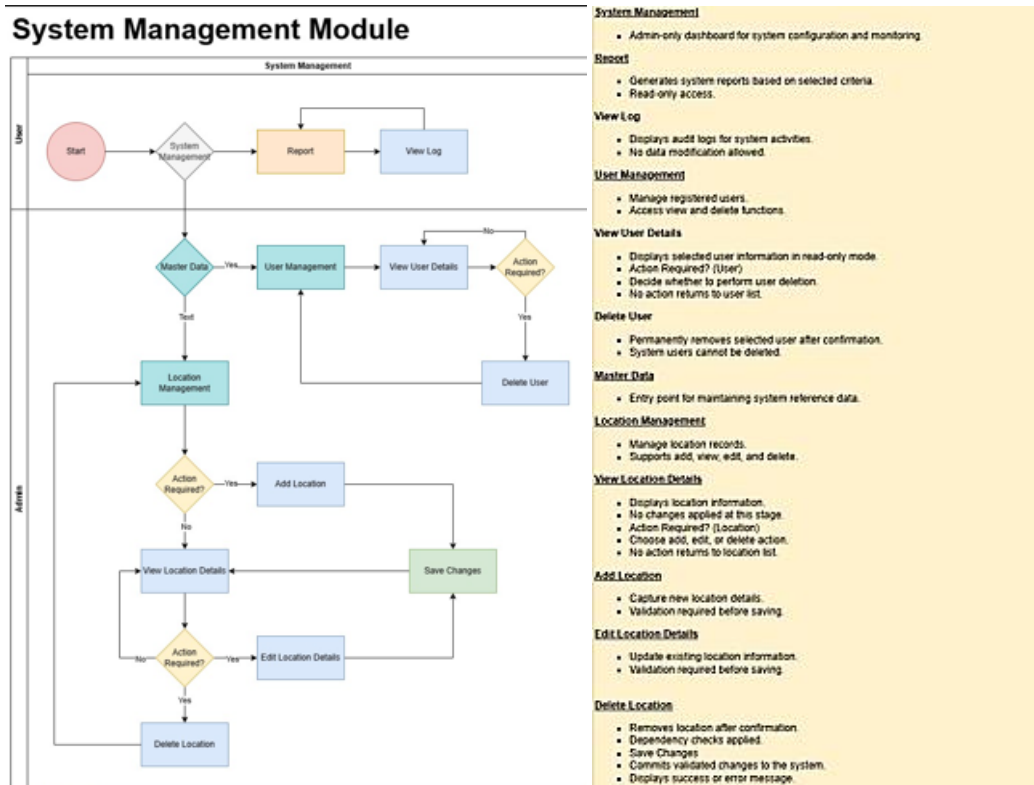


Image 4: UML Diagram for system management module

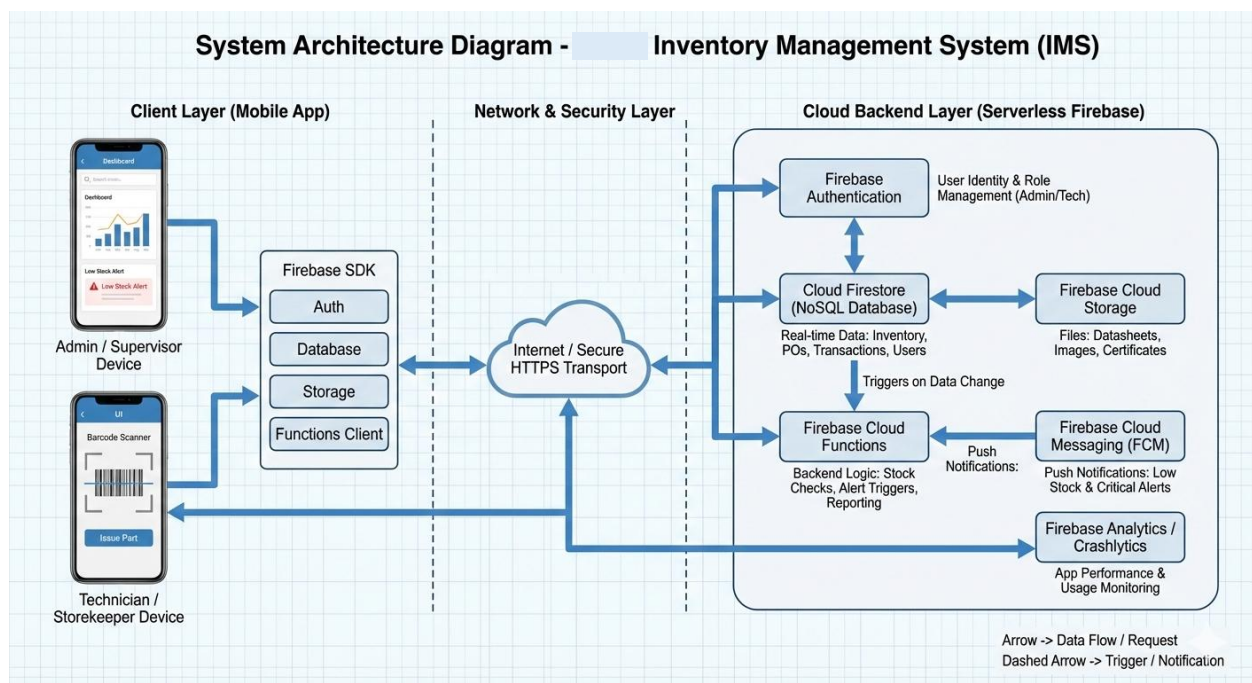


Image 5: System Architecture diagram

System Flow and Navigation Structure

The system flow begins with user authentication, followed by a role-based dashboard. Users can navigate to inventory management, consumption tracking, purchase order handling, and reporting modules based on their assigned roles.

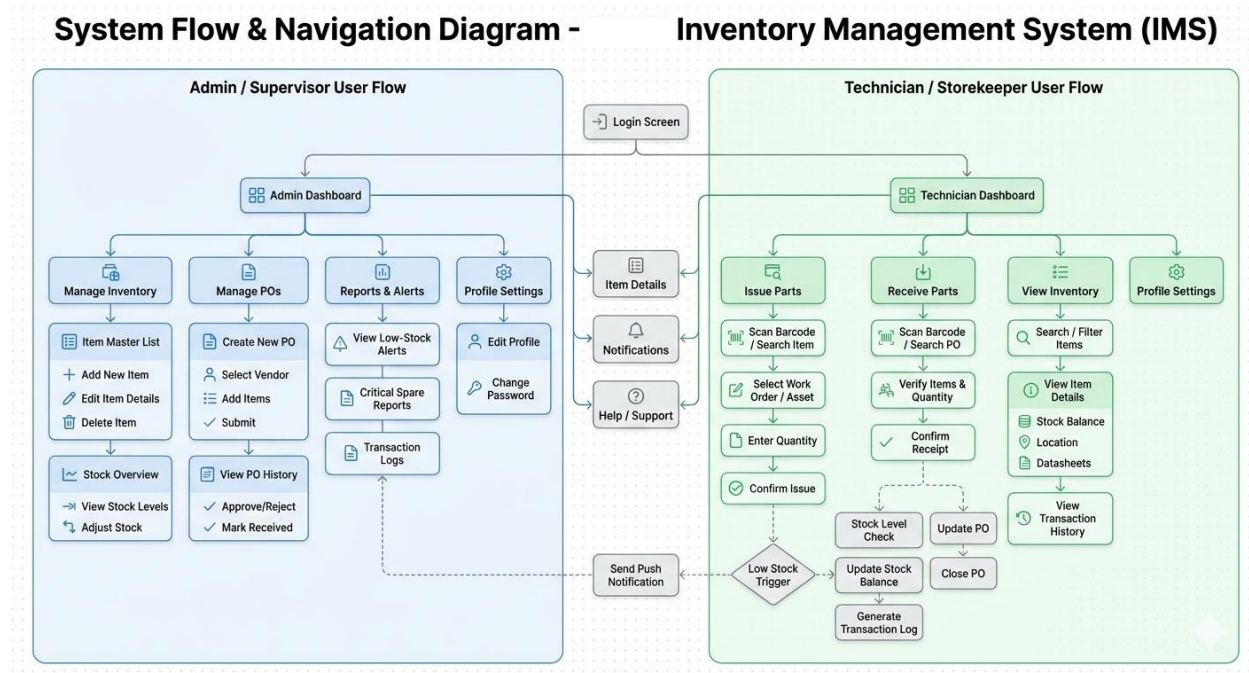


Image 6: System Flow and navigation diagram

Prototype Development Overview

This project includes a system prototype developed to demonstrate the overall user interface design, navigation structure, and logical workflows of an Inventory Management System. The prototype was designed with an emphasis on usability, clarity, and user experience rather than full technical or backend implementation.

Through the prototype, key system modules and screens are presented to illustrate how users navigate between functions such as inventory viewing, item management, and stock updates. It also demonstrates how user actions trigger logical system flows, helping users understand how different components of the system are interconnected. The prototype serves as a conceptual model to visualise system behaviour and user interaction in a practical and intuitive manner.

User Interface Design

The user interface design focuses on creating a clear, intuitive, and user-friendly interaction between users and the Inventory Management System. Special attention is given to simplicity and consistency to ensure that users with minimal technical background can operate the system efficiently. The interface design supports the project's purpose of improving inventory tracking accuracy and reducing operational errors within the engineering store.

Low-Fidelity Design

Low-fidelity wireframes are used to illustrate the basic layout, structure, and navigation flow of the system screens without detailed visual elements. These wireframes focus on functional placement, screen hierarchy, and user navigation paths. By using low-fidelity designs, early feedback can be gathered on usability and workflow efficiency, ensuring that the system structure aligns with project requirements before moving to more detailed design stages.



Image 7: Low Fidelity wireframe of the Inventory Management System

High-Fidelity Design

High-fidelity designs represent the final visual appearance of the system prototype, incorporating visual elements such as colours, icons, typography, and layout consistency. These designs aim to simulate a realistic user experience and demonstrate how the system would appear in a real deployment environment. The high-fidelity prototype helps validate usability, readability, and visual clarity, supporting the project goal of creating a practical and user-centred inventory management solution.



Image 8: High Fidelity wireframe of the Inventory Management System

Screen Context Explanation

Home Screen

The Home Screen serves as the primary entry point to the Inventory Management System. It provides users with an overview of system functionality and quick access to frequently used modules. By centralizing key functions, the Home Screen improves navigation efficiency and reduces the time required for users to perform routine inventory tasks, directly supporting operational efficiency.

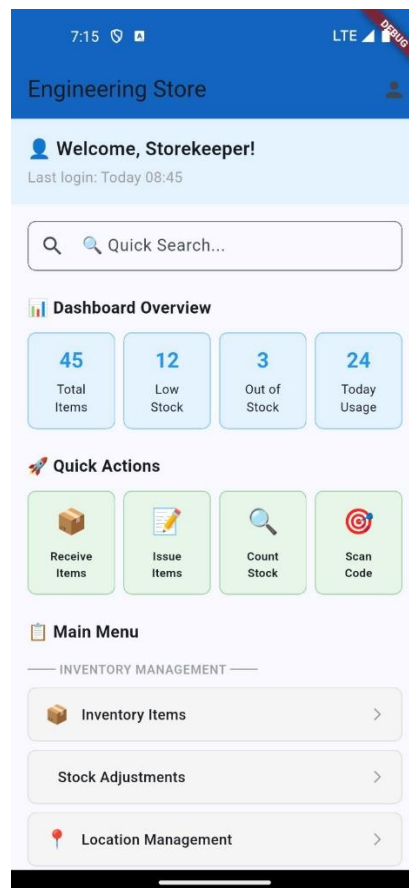


Image 9: Home Screen of the Inventory Management System

Menu Listing

The Menu Listing screen displays all available system modules in a structured and organized manner. This design allows users to easily identify and access different system functions without confusion. Clear module categorization enhances usability and ensures smooth navigation across the system, aligning with the project's objective of simplifying inventory-related operations.

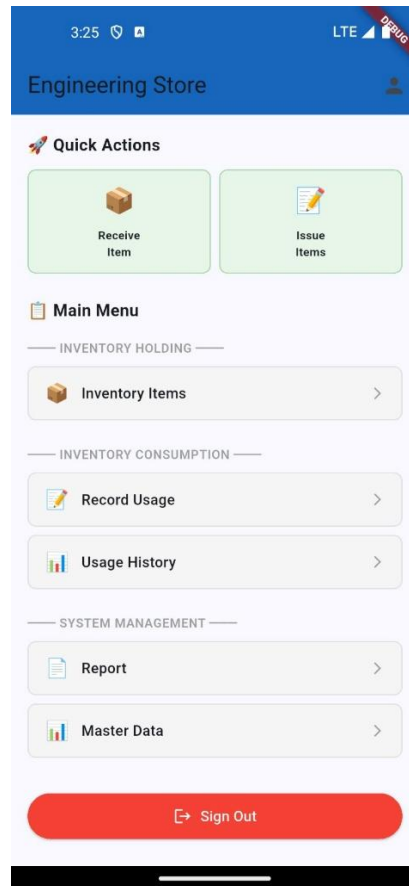


Image 10: Main Menu screen of the Inventory Management System

Add Item Screen

The Add Item screen simulates the process of recording new inventory items into the system. Users are guided to input relevant information such as item name, quantity, and category before confirming the data entry. This screen demonstrates how the system helps standardize inventory records, reduce manual entry errors, and improve data consistency within the engineering store.

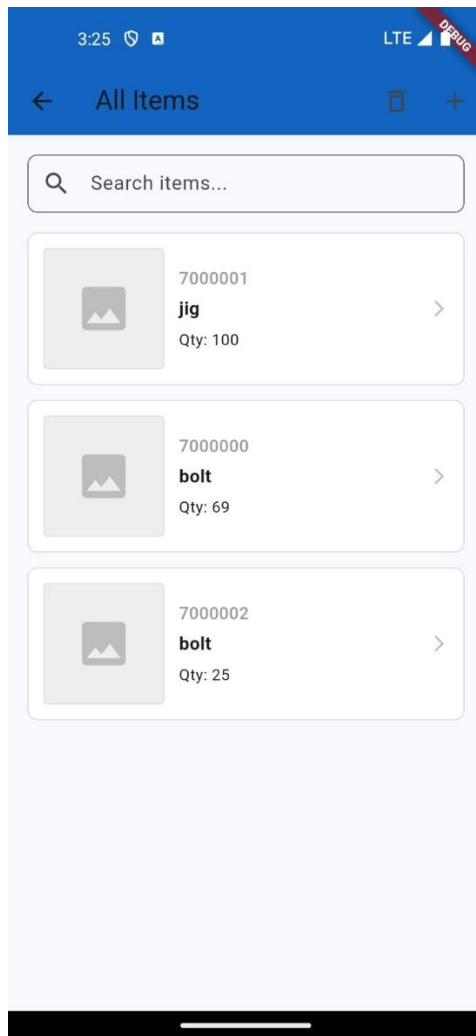


Image 7: Add or search items screen of the Inventory Management System

Requirement Validation

System requirements are validated through alignment with project objectives, stakeholder feedback, and consistency with UML diagrams and defined system workflows. This validation process ensures that the prototype accurately reflects functional requirements and logical system behavior. By validating requirements at the prototype stage, potential design issues can be identified early, supporting the project's goal of developing an effective and reliable inventory management solution.

Academic Justification

The development of a system prototype aligns with standard software engineering practices by enabling early requirement validation, usability evaluation, and workflow visualization. Prototyping reduces development risks by identifying potential issues before full system implementation. From an academic perspective, this approach demonstrates the application of theoretical concepts such as user-centered design, system modeling, and requirement analysis within a practical project context.

Conclusion

The development of a system prototype aligns with standard software engineering practices by enabling early requirement validation, usability evaluation, and workflow visualization. Prototyping reduces development risks by identifying potential issues before full system implementation. From an academic perspective, this approach demonstrates the application of theoretical concepts such as user-centered design, system modeling, and requirement analysis within a practical project context.